Code No: 157EN **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
ELECTRONIC SENSORS
(Common to CE, EEE, ME, CSE, EIE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Give the classification of transducers. Explain the basic requirements of a transducer.
b) Distinguish between precision and accuracy. [8+7]

2.a) Explain about stress sensors using quartz resonators.
b) Define the sensitivity and linearity of sensor and also write the equations for both. [7+8]

3. What is a Thermometer? Name different types of Thermometers and explain the working principle of all types of Thermometer in detail. [15]

4.a) Discuss about the Dielectric Constant and Refractive Index of thermo sensors.
b) Write a short note on Heat Flux Sensors. [9+6]

5.a) Discuss about working principle of Semiconductor Magneto resistors.
b) Explain the principle of operation of Eddy Current Sensors. Also, state its advantages and disadvantage. [7+8]

6.a) Write the applications of magnetic sensors in various field.
b) Explain about Angular/Rotary Movement Transducers with its applications. [6+9]

7.a) Discuss the basic characteristics of radiation sensors.
b) Explain about the radiation sensor? How does a radiation sensor works? [7+8]

8.a) Explain the working principle of Aerospace Sensors and also write the applications of few sensors.
b) Name the different primary sensors and explain about any two primary sensors. [7+8]

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